

Data Intake Processing and Verification Report

Background Information

- **Original Dataset Name:** The U.S. Greenhouse Gas and Air Pollutant Emissions System (GRA2PES)
- **GHG Center Dataset Title:** GRA²PES Greenhouse Gas and Air Quality Species
- **Dataset Provider:** NIST, NOAA
- **Date Obtained:** August 2024
- **Location Obtained From:** N/A - direct delivery to US GHG Center S3 bucket (source data can be obtained from <https://doi.org/10.18434/mds2-3520>)
- **Data Location in GHG Center:** gra2pes-ghg-monthgrid-v1
- **Data POC(s):** Dr. Brian McDonald, Dr. Meng Li
- **Dataset File Type(s):** NetCDF
- **Projection (if different from WGS84):** N/A

Data Transfer Confirmation

An SHA-256 checksum is used to detect high-level errors within data transmissions. Results from individual checksum file comparisons of pre-transfer and post-transfer shows all files were transferred successfully and no individual files had any transfer issues.

Data Intake Process

- https://us-ghg-center.github.io/ghgc-docs/data_workflow/gra2pes-ghg-monthgrid-v1_Data_Flow.html

Overall Dataset Statistics

- Statistics across all files:

	Minimum (tonne/km2/month)	Maximum (tonne/km2/month)	Mean (tonne/km2/month)	Standard Deviation
Original Data	0.0	110011.766	5.175	172.271
Transformed Data	0.0	110011.766	5.175	172.271

Monthly Dataset Statistics for Total Fossil Fuel emissions

- Statistics for Total Fossil Fuel CO₂ Emissions in 2021-03:

	Minimum (tonne CO ₂ /km ² /month)	Maximum (tonne CO ₂ /km ² /mont h)	Mean (tonne CO ₂ /km ² /month)	Standard Deviation
Original Data	0.0	99142.828	25.646	354.345
Transformed Data	0.0	99142.828	25.646	354.345

- Statistics for Total Fossil Fuel NO_x Emissions in 2021-09:

	Minimum (tonne NO _x /km ² /mont h)	Maximum (tonne NO _x /km ² /month)	Mean (tonne NO _x /km ² /mont h)	Standard Deviation
Original Data	0.0	144.753	0.067	0.518
Transformed Data	0.0	144.753	0.067	0.518

Visual Comparison:

Original Data for Total Fossil Fuel CO₂ Emissions 2021-01 visualized on QGIS



Transformed Data Total Fossil Fuel CO₂ Emissions 2021-01 visualized on QGIS



- Link to transformation record in [Jupyter Notebook](#)
- All values are in expected range

Summary

- We are confident that the transformation and display of data in the GHG Center is accurate
- [Link](#) to the transformation code
- [Link](#) to the dataset Tutorial

- [Link](#) to the US GHG Center data catalog page

Report Completed on: 8/20/2024

MSFC POC for questions: [Jeanné le Roux](#), [Siddharth Chaudhary](#)